DE–6874 **11**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

ELECTRONIC DEVICES AND DIGITAL CIRCUITS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) What are the advantages of Grey Codes? Discuss their areas of applications. (6)

   (b) Subtract using Excess – 3 code :

   (i) 336 – 241.

   (ii) 425 – 326. (4)

   (c) Add the following binary numbers.

   (i) 1100 + 1001

   (ii) 101.011 + 11.110

   (iii) 1000.011 + 1011.1010

   (iv) 1011001 + 11010. (10)

2. (a) Explain the logic diagram of arithmetic circuits. (6)

   (b) Convert the following :

   (i) $(5A)_{16} = (\ ?\ )_{10}$

   (ii) $(3A9)_{16} = (\ ?\ )_{10}$

   (iii) $(950)_{10} = (\ ?\ )_{16}$

(iv) $(428)_{10} = (\ ?\ )_{16}$

(v) $(6BC)_{16} = (\ ?\ )_{2}$

(vi) $(435)_{8} = (\ ?\ )_{10}$

(vii) $(111011)_{2} = (\ ?\ )_{10}$. $(7 \times 2 = 14)$

3. (a) Explain how the NAND and NOR gates can act as universal building blocks. (15)

   (b) State and explain De Morgan's theorem. (5)

4. (a) What are the basic laws of Boolean algebra? Discuss. (12)

   (b) Explain the working of XOR gate and how it is used for binary applications. (8)

5. (a) What is a full-adder? Write the truth table for a full-adder and develop its logic network. (14)

   (b) Explain the working of a half subtractor. (6)

6. Minimize the following function using the Quine Mc Cluskey method.

$$f(x_1, x_2, x_3, x_4, x_5) = \sum (0,1,2,3,5,7,9,11,17,19,29.31) + \sum_{d} (4,6,13,15,21,23,25,27)$$

7. (a) With a neat block diagram discuss the working of a J-K flip flop. (10)

   (b) Discuss the working principle of a synchronous counter with its block diagram. (10)

8. (a) Discuss in detail about the fundamentals of BJT operation. (12)

(b) Give a short note about the

(i) Silicon control rectifier. (4)

(ii) Unijunction transistor. (4)

DE–6875 **12**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks

($5 \times 20 = 100$)

1. (a) Explain briefly about the various developments in computer technology. (10)

   (b) Explain briefly about the various developments in communications technology. (10)

2. Explain in detail about word processing, spreadsheet and database software.

3. (a) Explain about video conferencing and picture phones. (10)

   (b) Write about Electronic Data Interchange and Intranets. (10)

4. (a) Write about compression and decompression. (10)

   (b) Explain the features of a Database Management System. (10)

5. (a) Explain about the six phases of system analysis and design. (15)

   (b) Describe the features of HTML. (5)

6. (a) Write in detail about Local Area Networks. (15)

   (b) Describe the factors that will affect the data transmission. (5)

7. Write about

   (a) Hard Disks. (7)

   (b) Optical Disks (7)

   (c) Magnetic Tapes. (6)

8. (a) Explain the five steps in the programming process. (15)

   (b) Write the benefits of object oriented programming. (5)

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DE–6876 **13**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

PRINCIPLES OF OPERATING SYSTEM

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Discuss the history of operating systems.

   (b) What is an operating system? Explain it.

2. (a) Write a detailed note on Interprocess communication.

   (b) Explain round robin scheduling.

3. (a) Discuss the goals of I/O software.

   (b) Explain in detail about deadlock modeling.

4. (a) Discuss multiprogramming with fixed partitions.

   (b) Write a short note on paging.

5. (a) Write a detailed note on directory structure.

   (b) Explain about concurrency control and transactions.

6. (a) Describe the FIFO page replacement algorithm.

   (b) Write a note on user Authentication.

7. (a) Give a note on Semaphores.

   (b) Explain about device controllers.

8. (a) Discuss in detail about file storage.

   (b) Explain the design principles for security.

DE–6877 **14**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

C PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Explain data types in C with examples. (10)

   (b) What do you mean by top–down approach? Explain with an example. (10)

2. (a) Explain in detail the concept of C tokens. (10)

   (b) List out the operators available in C. Explain any two operators with suitable example. (10)

3. (a) Write short on

      (i) Nested loops. (5)

      (ii) The goto statement. (5)

   (b) What is an array? How do you declare and initialize arrays? (10)

4. (a) What do you mean by user–defined functions? Why we need user–defined functions? (10)

   (b) Explain the concept of functions with arrays. Give an example. (10)

5. (a) Explain the concept of dynamic storage allocation with an example program. (10)

   (b) Create a employee structure with the members employee id, name, basic salary, TA, DA, HRA for ten employees and also calculate the net salary of each employee and display them. (10)

6. (a) How to open and close a file? Give an example. (10)

   (b) Explain the concept of random access operation in file handling with example. (10)

7. (a) (i) Write short notes on command line argument. (5)

   (ii) Write the differences between structure and union. (5)

   (b) Write a C program which calls a function reverse () which accepts a string and display its reverse. (10)

8. (a) Write a C program using pointers to find the biggest of two given numbers. (10)

   (b) Discuss about data types of C. (10)

DE–6878 **15**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

ACCOUNTING FUNDAMENTALS

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Define Accounting. Explain its scope and functions. (10)

   (b) Explain the managerial uses of Management Accounting. (10)

2. Briefly explain various accounting concepts and conventions.

3. (a) What are Subsidiary Books? Explain its kinds. (10)

   (b) Distinguish between Trial Balance and Balance Sheet. (10)

4. (a) Explain the various types of ratio. (10)

   (b) Write detailed note Inter-Firm and Intra-Firm comparison. (10)

5. The following are the balances extracted from the books of Ganesh as on 31.12.2006. Prepare Trading and Profit and Loss account for the year ending 31.12.2006 and a Balance Sheet as on that date.

Trial Balance as on 31.12.2006

Debit Balances	Rs.	Credit Balances	Rs.
Drawings	4,000	Capital	20,000
Cash at Bank	1,700	Sales	16,000
Cash in hand	6,500	Sundry creditors	4,500
Wages	1,000		
Purchases	2,000		
Stock 1.1.2006	6,000		
Buildings	10,000		
Sundry debtors	4,400		
Bills receivable	2,900		
Rent	450		
Commission	250		
General Expenses	800		
Furniture	500		
	40,500		40,500

The following adjustments are to be made :

(a) Stock on 31.12.2006 was Rs. 4,000.

(b) Interest on capital at 6% to be provided.

(c) Interest on drawings at 5% to be provided.

(d) Wages yet to be paid Rs. 100.

(e) Rent prepaid Rs. 50.

6. The trial balance of a book keeper shows an excess of debits over credits by Rs. 261. This difference is placed in a suspense account to facilitate closing of the books. Later on the following errors were discovered. Rectify the errors and prepare account at Rs. 439.
   (a) A credit item of Rs. 349 has been debited to personal account at Rs. 439.
   (b) A sum of Rs. 625 written off fixtures as depreciation has not been posted to depreciation account.
   (c) Rs. 9,000 paid for furniture bought have been charged to the purchase account.
   (d) A discount allowed to a customer has been credited to him as Rs. 145 in place of 154.
   (e) A sale of Rs. 594 was posted as Rs. 495 in sales account.
   (f) The total of Returns Inward Book has been added Rs. 10 short.

7. Hari maintain his books of account on single entry system. His books provide the following information :

	1.4.2005	31.3.2005
	Rs.	Rs.
Furniture	200	200
Stock	2,800	3,050
Debtors	2,100	3,400
Cash	150	200
Creditors	1,750	1,900
Bills receivable	—	300
Loan taken	—	500
Investments	—	1,000

His drawings the years were Rs.500. Depreciate furniture by 10% and make a provision for doubtful debts @ 10% on debtors. Prepare the statement showing profits for the year.

8. From the following information make out a statement of proprietors funds with as many details as possible :

(a) Current ratio 2.5

(b) Liquidity ratio 1.5

(c) Proprietary ratio (fixed assets/proprietary fund) 0.75

(d) Working capital Rs. 60,000.

(e) Reserves and Surplus Rs. 40,000.

(f) Bank overdraft Rs. 10,000.

(g) There is no long-term loan or fictitious asset.

DE–6879 **21**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

MICROPROCESSORS AND PERSONAL COMPUTERS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What is the basic difference between the microprocessor and the micro computer? Explain. (5)

   (b) What are the advantages of microprocessor based system? Discuss. (5)

   (c) What is bus? Explain the various buses and their functions in a microprocessor system. (10)

2. (a) With suitable diagram, explain the architecture of 8085 microprocessor? (10)

   (b) Explain the various registers in 8085. (10)

3. (a) What are the addressing modes available in 8085? Explain. (10)

   (b) Explain the various Data transfer instructions in 8085 microprocessor. (10)

4. (a) Write an assembly language program to add two numbers of 16 bit data stored in memory from 4200_H to 4203_H. (10)

(b) Write an assembly language program to find the multiplication of two 8 bit data stored in memory 4200_H and 4201_H and store the product in 4202_H and 4203_H. (10)

5. (a) What is ROM? Explain the different types of ROM. (10)

(b) Give a brief note about the memory Hierarchy. (10)

6. (a) Describe in detail about the Isolated I/O and memory mapped I/O methods. (10)

(b) What is DMA? Explain. (10)

7. (a) Explain IDE and SCSI drive controller. (10)

(b) Write short note about the video cards. (10)

8. (a) Discuss in detail about the importance of preventive maintenance. (10)

(b) Explain about the data backup system. (10)

DE–6880 **22**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

OFFICE AUTOMATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) List and explain the purpose of Title Bar and status bar. (10)

   (b) Why we need document? How will you closing, deleting and undeleting a document? (10)

2. (a) What is the purpose of paragraph? Explain in detail. (10)

   (b) How do you create a table? How can you format it? (10)

3. (a) What are the arithmetic functions used in Excel? Explain. (10)

   (b) Explain about creating, enhancing and printing a chart. (10)

4. (a) How do you create a database? How will you create and save a table? (10)

   (b) Explain in detail about forms in Access. (10)

5. (a) How can you create an access report from an Excel list? (10)

   (b) Discuss how to use menus and dialog boxes in outlook window. (10)

6. (a) Give short notes on pop-up menus in windows. (10)

   (b) What is the purpose of creating folders and shortcuts in windows? (10)

7. (a) Explain different types of Table Auto Format in Excel. (10)

   (b) Write the steps for creating employee database and manipulate with records. (10)

8. (a) Discuss on special effects in power point presentation. (10)

   (b) Explain about drawing toolbar in word. (10)

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DE–6881 **23**

DISTANCE EDUCATION

B.C.A/B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013

SYSTEMS ANALYSIS AND DESIGN

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What are the elements of a system? Explain.
   (b) Explain the role of systems analyst.

2. (a) How to conduct the investigation? Discuss.
   (b) Explain the fact finding techniques.

3. (a) Discuss the cost benefit analysis in detail.
   (b) How to analyse systems data? Discuss.

4. (a) Write a note on structured analysis.
   (b) Discuss about data flow diagram.

5. (a) Explain the input design of on–line systems.
   (b) Explain the stages of system design.

6. (a) Explain the types of files.
   (b) Explain the types of databases.

7. (a) Discuss in detail about warnier orr diagram.

   (b) Write a detailed note on system testing.

8. (a) Discuss about the conversion methods.

   (b) Give a note on system maintenance.

DE–6882 **24**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

OBJECT ORIENTED PROGRAMMING AND C++

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Explain the basic concepts of object oriented programming and their advantages. (10)

   (b) Explain the various types of C++ operators. (10)

2. (a) Define friend function. Explain with an example. (10)

   (b) Explain inline function with an example. (10)

3. (a) Define pointer. Explain the use of new and delete operators. (8)

   (b) Explain copy and dynamic constructor with example. (12)

4. (a) What is operator overloading? Explain with an example. (10)

   (b) Write a C++ program for adding different types of data using function overloading. (10)

5. (a) What is the use of inheritance? Explain the different types of inheritance with example. (12)

   (b) Define pure virtual function. Give an example. (8)

6. (a) Explain static and dynamic binding with example. (10)

(b) Write a C++ program for adding odd and even in a given set of numbers. (10)

7. (a) Explain the following with example : (8)

(i) Public

(ii) Private

(iii) Protect

(b) Write short notes on : (12)

(i) This pointer

(ii) Stream classes

(iii) Static data

8. (a) Explain the following : (10)

(i) Abstract class

(ii) Virtual base class.

(b) Write short notes on : (10)

(i) Data encapsulation

(ii) Polymorphism.

DE–6883 **25**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

COMPUTER GRAPHICS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) What is meant by the interactive Computer Graphics? List and explain the functions of a computer graphics package. (10)

   (b) Explain the simple raster display system. (10)

2. What are the various input devices used for Graphics? Explain any four of them. (20)

3. (a) Explain the steps in Bresenham's line drawing algorithm. (10)

   (b) Describe the different algorithms used for region filling. (10)

4. Explain viewing transformation and discuss the role of clipping in the applications of these transformations. Describe any one 2-D clipping algorithm for lines. (20)

5. (a) What is the need of hidden-surface algorithms? How does the Z-buffer algorithm determine which surface are hidden? (10)

   (b) What are the basic two dimensional transformations? Explain with examples. (10)

6. (a) Explain the depth sort algorithm for hidden surface removal. (10)

   (b) Discuss about the Scan-line algorithm for hidden surface removal. (10)

7. (a) Write a brief note on shearing and reflection. (10)

   (b) Explain Sutherland-Hodgeman algorithm for polygon clipping. (10)

8. (a) Discuss alternatives for feedback in terms of the various levels of user. (10)

   (b) Explain any two popular graphical user interfaces. (10)

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DE–6884 **31**

DISTANCE EDUCATION

B.C.A./ B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

BUSINESS COMMUNICATION

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. ($5 \times 20 = 100$)

1. (a) Briefly explain the goals of communication. (10)

   (b) What are the common barriers to communication in an organisation? (10)

2. (a) State the essentials of a good business letter. (10)

   (b) Write a letter on behalf of satya furnitures, Chennai asking for quotation of prices for their computer tables and chairs. (10)

3. (a) What are the points to be considered to draft complaints letter? (10)

   (b) Write a letter to Williams Ltd, Mumbai complaining about the chocolates sent by the company are of poor quality. (10)

4. (a) What are the characteristics of a good report? (10)

   (b) You have recently joined as a branch manager of a company. Write a report to the head office on the defects in the routine work of the branch. (10)

5. Draft the minutes of the statutory meeting of a company.

6. (a) Briefly discuss the steps in planning a talk. (10)

   (b) Draft a speech to mourn the death of a colleague who has been killed in an accident. (10)

7. Write an essay on "Environmental pollution –causes and Remedies".

8. (a) What is Biodate? Write the contents of Biodata. (10)

   (b) VIJAY & CO., has advertised in the Thina Thandi for the post of supervisor in sales department. With reference to this advertisement write an application. (10)

DE–6885 **32**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

WINDOWS AND VISUAL BASIC

(2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) What is GUI? Explain in detail. (10)

   (b) Explain how to Select, Move and Size the window. (10)

2. (a) What is the purpose of clipboard? Explain. (10)

   (b) What do you mean by OLE? Explain it. (10)

3. (a) Write the important properties and methods of Text Box control. (10)

   (b) Write a VB program to change the range of value using scroll bars. (10)

4. (a) Discuss about Toolbox in VB. (10)

   (b) Write short note on:

      (i) If .... End If.

      (ii) Select case statement. (10)

5. (a) What do you mean by procedure? Explain the types of procedures. (10)

   (b) Write the steps for creating Inventory control using MDI forms. (10)

6. (a) Discuss about Menus of windows. (10)

   (b) Give brief notes on Print Manager. (10)

7. (a) How do you create menus using Menu Editor? (10)

   (b) How to make EXE file? Explain. (10)

8. (a) Write short notes on:

   (i) MSG Box

   (ii) Input Box.

   (b) Write a VB program to create a calculator using Control Array.

DE–6886 **33**

DISTANCE EDUCATION

B.C.A./B.C.A.(Lateral) DEGREE EXAMINATION,
DECEMBER 2013.

INTERNET AND JAVA PROGRAMMING

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions

(5 × 20 = 100)

1. (a) Discuss about Client Server Model. (10)

   (b) Discuss about TCP/IP protocols. (10)

2. (a) What do you mean by Domain Name System? Explain. (10)

   (b) Discuss about E-mail. (10)

3. (a) Discuss about File Transfer Protocol. (10)

   (b) Define and explain Fragmentation and Routing. (10)

4. (a) Explain Web Server and Web Browser. (10)

   (b) Compare and contrast FTP with HTTP (10)

5. (a) Explain Internet resources (10)

   (b) Briefly explain about Entertainment on Internet(10)

6. (a) Explain in detail about data types in Java. (10)

   (b) Write notes on Object Oriented Programming principles. (10)

7. (a) Discuss in detail about I/O handling in Java. (10)

   (b) How will you create an Applet? (10)

8. (a) Explain Constructor with example. (10)

   (b) Explain about Exception in Java with suitable example. (10)

DE–6887 **34**

DISTANCE EDUCATION

B.C.A/B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

RDBMS

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Write notes on Database Schema. (10)

   (b) Describe the Data Manipulative capabilities in detail. (10)

2. State and explain the difficulties of Relational Model. (20)

3. (a) What do you mean by Data Manipulation Facilities? Explain. (10)

   (b) Briefly explain the Data Control facilities of a Relational Model. (10)

4. (a) Explain the Properties of Relational Database. (10)

   (b) Write notes on Oracle architecture. (10)

5. (a) Briefly explain the Data Access SQL Commands. (10)

   (b) What are DML commands? Explain with examples. (10)

6. Write SQL queries to

   (a) Create a Table Called "STUDENT" with the fields like Reg.No., Name, Total, Grade, Result. (7)

   (b) Display all the Students detail. (4)

   (c) Display all the students records whose result is pass. (4)

   (d) Display all the records for total marks between 60 and 80. (5)

7. Explain the Join operation by creating tables of your own. (20)

8. (a) Briefly explain about the database objects. (10)

   (b) Write notes on union and multiple-part quires. (10)

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DE–6888 **35**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

MANAGEMENT PRINCIPLES AND TECHNIQUES

(2003 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. Explain the concepts, scope and principles of management system as applicable to business and Industry. (20)

2. Define staffing and Delegation. Discuss about the various dilutions and motivations in management principles. (20)

3. Discuss the historical backgrounds of operation research. Explain its significance and scope in modern management. Enumerate the limitations of operation research. (20)

4. Solve the following LP problem by Simplex method.

   Minimize $z = x_1 + -3x_2 + 2x_3$

   Subject to $3x_1 - x_2 + 3x_3 \leq 7$

   $-2x_1 + 4x_2 \leq 12$

   $-4x_1 + 3x_2 + 5x_3 \leq 10$

   $x_1;\ x_2;$ and $x_3 \geq 0$. (20)

5. A project has the following time schedule.

Activity	Time in weeks	Activity	Time in weeks
(1-2)	4	(5-7)	8
(1-3)	1	(6-8)	1
(2-4)	1	(7-8)	2
(3-4)	1	(8-9)	1
(3-5)	6	(8-10)	8
(4-9)	5	(9-10)	7
(5-6)	4		

Construct a PERT network and compute.

(a) T_E and T_L for each event

(b) Float for each activity

(c) Critical path and its duration. (20)

6. What is meant by resource levelling? Discuss about the cost consideration of project scheduling with suitable example. (20)

7. The cost of Machine is Rs. 6100 and its scarp value is only Rs.100. The maintenance cost are found from experience to be.

Year:	1	2	3	4	5	6	7	8
Maintenance cost in Rs.	100	250	400	600	900	1250	1600	2000

When should machine should be replaced? (20)

8. Explain how the theory of replacement is used in the following problem.

   (a) Replacement and items whose maintenance cost varies with time.

   (b) Replacement and items that fail completely. (20)

DE-7619 **36**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

LAB III — JAVA AND VISUAL BASIC
(2003 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a VB program to display the day of the given date. Also convert the lower case to upper case for the given word.

   (b) Write a Java program to find simple and compound interests using class concept.

---------- Cut here ----------

2. (a) Write a VB program to select candidates for four posts based on their qualifications :

Male and PG Degree	:	Manager
Male, Graduate and Typing	:	Clerk
Female and PG Degree	:	Secretary
Female, Graduate and Typing	:	Steno

   (b) Write a Java program to sort the given set of numbers in ascending order.

3. (a) Write and test a VB program to print twenty address labels in a two column format for the address :

The Registrar

Alagappa University

Karaikudi – 630 003.

(b) Write a Java program to find the factorial of a given number using class concept.

---------- Cut here ----------

4. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

(b) Write a Java program to find square root, square and cube of any number.

---------- Cut here ----------

5. (a) Write and test a VB program to compute the surface area and volume of a sphere given the radius. Use option button and input boxes.

(b) Write a Java program to input a temperature in celsius and find the corresponding temperature in fahrenheit.

---------- Cut here ----------

6. (a) Write a test a VB program to compute and print either the sum or the product of the first N natural numbers. Use option button.

(b) Write a Java program to accept two strings. Assign a string to another variable. Concatenate the two strings and find the length of the concatenated string.

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DE–7617 **16**

DISTANCE EDUCATION

B.C.A. DEGREE EXAMINATION, DECEMBER 2013.

LAB I — C PROGRAMMING

(2003 onwards)

Time : Three Hours Maximum : 100 Marks

Examiner has to select and give ONE question to each candidate by lot system.

1. (a) Write a C program that receives the data such as age and name of person to check the eligibility for voting. Take the condition that if a person is more than 18 years old he is eligible to vote. Else display the number of years, he has to wait for voting.

   (b) Write a C program to find a word is palindrome or not.

---------- Cut here ----------

2. (a) Write a C program to sort the given set of numbers in ascending order.

   (b) Write a C program to accept and display your date of birth using structure.

---------- Cut here ----------

3. (a) Write a C program to find simple and compound interests using function.

   (b) Write a C program to read the records in an address file.

4. (a) Write a C program to compute commission earned by a salesman according to the scheme given below:

Sales Amount	Commission
Upto 10,000	5%
Above 10,000 to 50,000	10%
Above 50,000	20%

(b) Write a C program to create a text file and display the contents of that file.

---------- Cut here ----------

5. (a) Write a C program to find the maximum and minimum in a given set of numbers.

(b) Write a C program to display the address and the contents of pointer variable.

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6. (a) Write a C program to arrange the names in alphabetical order.

(b) Write a C program to find a bigger number between the two numbers using pointer method.

DE–7618 **26**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

LAB II — C++ AND MS OFFICE

(2003 onwards)

Time : Three hours Maximum : 100 marks

1. Examiner has to select ONE question for each candidate by lot system.
2. Each subdivision carry 50 marks.
3. Write an algorithm or flowchart for C++.
4. Write the procedure and draw screenshot for MS Office.

1. (a) Create a class called EMPLOYEE that contains employee number, employee name, designation, basic pay and deductions (LIC & PF). Include member function to get data from user for n employees. Write a C++ program to prepare the payslips for n number of employees using the following details :

HRA = 400; DA = 40% of basic pay.

Gross pay = Basic pay + DA + HRA

Net pay = Gross pay - Deductions.

Print the result in the following format.

Emp No.	Emp name	Basic	LIC	PF	DA	HRA	Gross pay	Net pay
_	–	–	–	–	–	–	–	–

(b) Using the MS-Word type out the title page of your OFFICE AUTOMATION book.

Do the following formatting features :

(i) (1) Centre each line

(2) Choose appropriate font size in each line

(3) Choose font type.

(ii) Draw border lines.

(iii) Use colour options.

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2. (a) Write a C++ program to create a class of objects e1 and e2. The contents of object e1 is added to the object e2 using the operator overloading technique.

(b) Create the following worksheet in EXCEL :

REG NO.	NAME	MATHS	ENGLISH	SCIENCE	TAMIL	TOTAL
_	_	_	_	_	_	_
_	_	_	_	_	_	_

(i) Fill the TOTAL column

(ii) Fill the candidates whose REG. NO. > = 100

(iii) Filter the candidate whose NAME = 'BALU'

(iv) Filter the candidate whose MATHS marks > 80.

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3. (a) Write a C++ program to find the square of a given number with different arguments using function overloading.

(b) Create LIBRARY table with the following fields :

Book Number, Title, Author, Price, Publisher and Branch.

(i) Add 5 records using FORMS.

(ii) Display the records for Book Number = 101 AND Branch = 'Commerce'.

(iii) SORT the table in the ascending order of Book Number.

(iv) Display the records for publisher = 'PHI' OR Publisher = 'BPB' AND price between 400 AND 500.

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4. (a) Write a C++ program to read the derived class data members such as name, roll number, sex, height and weight from the key-board and display the contents of the class on the screen. (Using single inheritance concept)

(b) Create sales table with the following fields :

Name, sex, salescode and salesamount.

(i) Add 5 records using FORMS.

(ii) Display the records for salesamount > 1000 and sex = 'M'.

(iii) Display the records for sex = 'F' and salesamount between 5000 and 10000.

(iv) SORT the table in descending order of salesamount.

---------- Cut here ----------

5. (a) Write a C++ program to illustrate such as addition, subtraction, multiplication and division using a member function which are defined out of the scope of a class definition.

(b) Create a worksheet with the following fields :

Year, sales and expense.

(i) Draw the graph and mention the appropriate headings.

(ii) Include more columns (say profit) with data values and create different types of graphs.

6. (a) Write a C++ program to display the given set of numbers in ascending/descending order.

   (b) Using MS-Word type a circular letter to be posted to MCA candidates to attend PCP. Format the letter and apply suitable font type and font size.

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7. (a) Write a C++ program to read and display the students particulars such as Reg. No., name, age and sex using class and objects.

   (b) Create employee table with the following fields :

   Name, Eid, Designation, department and salary.

   (i) Add 6 records to the table.

   (ii) Display all the records who are working in the sales department.

   (iii) Display the name of all the employees whose name starts with s.

   (iv) Sort all the records in ascending order based on the field Eid.

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8. (a) Write a C++ program to find the mean and standard deviation of given set of numbers.

   (b) Create the worksheet with the following fields :

   S.No., Reg. No., Name, Age, Maths, Physics, Chemistry, Total, Average and result.

   (i) Find the total and average marks.

   (ii) Find the maximum and minimum mark for each subject.

   (iii) Replace the result with pass for average > = 50 else fail.

   (iv) Sort this list by Reg. No. field.

DE–7620 **36**

DISTANCE EDUCATION

B.C.A./B.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2013.

LAB III — ORACLE AND VISUAL BASIC LAB

(Common for all batches)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the Examiner and given to each candidate by lot.

Each subdivision carries 50 marks.

1. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

   (b) Create table MARK with the following fields :

   REG NO, NAME, TAMIL, ENGLISH, MATHS, BIOLOGY

   (i) Add 5 records

   (ii) Show data in fields NAME, REG. NO. & MATHS alone on the screen

   (iii) Display the ascending order list with NAME as primary key

   (iv) Display all records having TAMIL > 60.

2. (a) Write and test a VB program to display the day of the given date and convert upper case from lower case to given word.

(b) Create table ADDRESS with the following fields :

NAME, STREET, CITY, PIN

(i) Add 5 records

(ii) Display the structure of the table

(iii) Add the field DISTRICT

(iv) Fill the DISTRICT with 'CHENNAI'

(v) Change the PIN to 630003 where CITY is Karaikudi and PIN is 623003.

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3. (a) Write a test a VB program to select candidates for four posts, based on their qualifications :

Male & PG degree	Manager
Male, graduate and typing	Clerk
Female and PG Degree	Secretary
Female, graduate and typing	Steno

(b) Create table RESULT with the following fields :

REG NO, NAME, M1, M2, M3, TOTAL, RESULT

(i) Add 5 records

(ii) Fill TOTAL field with $M_1 + M_2 + M_3$

(iii) Fill the RESULT with 'pass' if TOTAL > = 150

Otherwise 'FAIL'

(iv) Display all records whose RESULT is PASS.

4. (a) Write a VG program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

(b) Create table SALES with the following fields :

SALENO, SALENAME, SALEAMT, COMM, SALEDATE

(i) Add 5 records

(ii) Replace COMM with the following conditions :

If SALEAMT < 1000	No Commission
SALEAMT ≥ 1000 & < 5000	5%
SALEAMT ≥ 5000	10%

(iii) Display all records

(iv) Display the records having COMM < 1000

(v) Display average sales and average commission.

---------- Cut here ----------

5. (a) Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text and enter key.

(b) Create table MARKS with the following fields :

REGNO, NAME, M1, M2, M3, TOTAL, AVG

(i) Add 5 records

(ii) Replace TOTAL with M1+ M2 + M3

(iii) Replace AVG with TOTAL/3

(iv) Count the number of records for M2 > 50.

6. (a) Write a VB program to compute and print either the sum of odd numbers or even numbers at the user's choice using label, text and option buttons.

(b) Create table NUMB with the following fields :

NUM, ABSNO, SQRTNO, ROUNDNO, SQRNO

(i) Add 5 records

(ii) Replace ABSNO with the absolute value of given number

(iii) Replace SQRTNO with square root value of given number

(iv) Replace SQRNO with square of given number

(v) Replace ROUNDNO by using ROUND option

(vi) Display all the records.